

Breathing In, Breathing Out — Mark schemes

PAPERS A TO D • 30 MARKS EACH • TUTOR COPY

Marks in square brackets show how the total is split. Almost every mark lost on this topic is lost to language rather than to knowledge, and always in the same three places: putting respiration in the lungs, giving a muscle the job that pressure does, and describing a feature without saying what it achieves. Three phrases are worth insisting on — "in every cell, in the mitochondria", "the volume changes, so the pressure changes, so the air moves", and "down the concentration gradient". The notes under each answer name the specific mistake that question was built to catch.

Paper A — Medium

30 MARKS

- 1 Diaphragm, breathing out: **relaxes and domes upwards** [6]
 Ribs, breathing in: **move up and out**
 Volume of thorax: breathing in **increases**; breathing out **decreases**
 Pressure in lungs, breathing out: **rises** (above atmospheric)
 Air, breathing in: **flows in**
One mark per box. "The diaphragm pushes up" is not the same as "relaxes" — relaxing is what earns the mark, and a student who writes push has the muscle doing something it cannot do.
- 2 (a) **Trachea** [4]
 (b) **Bronchi** (one bronchus per lung)
 (c) **Alveoli**
 (d) **Diaphragm**
Accept windpipe for trachea. Do not accept "lungs" for (c) — the question asks for the sacs.
- 3 Breathing (ventilation) is the movement of air into and out of the lungs, done by muscles [1]
 Respiration is a chemical reaction that releases energy from glucose, and it happens in every cell of the body, in the mitochondria [1]
The single most examined distinction in this topic. If the answer places respiration in the lungs, award nothing for the second mark however well it is written.
- 4 Any three, one mark each: **very large total surface area**; **wall only one cell thick** [3]
 (short diffusion distance); **moist lining**; **dense network of capillaries** / good blood supply maintaining the concentration gradient; **constantly ventilated**
"They are small" on its own scores nothing — smallness matters because of what it does to the surface area, and the answer has to say so.
- 5 (a) **Oxygen** [3]
 (b) **Carbon dioxide**
 (c) Both move by **diffusion, from where they are in higher concentration to where they are in lower concentration** — each down its own gradient
A common half-answer to (c) is "they both go through the wall". That is where, not why.
- 6 Gases must **dissolve in the liquid film** before they can diffuse across the cell membranes [1]
 so a dry surface would slow gas exchange down, or stop it [1]
Links directly to why a fish drowns in air — same idea, different animal.
- 7 (a) glucose + **oxygen** → **carbon dioxide** + water [2] (one mark per gap) [3]
 (b) **Mitochondria** [1]
If the student has added "energy" as a product in the equation, do not deduct here, but write "released, not produced" in the margin — it costs marks in worded questions later.

- 8 Limewater [1]** [2]
It turns **milky / cloudy**, and does so within a few breaths [1]
"It goes white" is acceptable; "it changes colour" alone is not specific enough.
- 9** Oxygen, expired: **16%** [3]
Carbon dioxide, inspired: **0.04%**
Nitrogen, expired: **78%**
The nitrogen box is the one that catches people out — many students change it because every other box changed. Nitrogen is not used by the body at all.
- 10** The **rate** of breathing increases — more breaths per minute [1] [2]
The **depth** of breathing increases — more air moved per breath [1]
Two marks, two distinct changes. "He breathes more" is one idea written twice.

Paper B — Medium

30 MARKS

- 1** (a) **Nitrogen [1]**; it is not used by the body in any reaction, so the same amount is [5]
breathed out as was breathed in [1]
(b) $4 \div 0.04 = \mathbf{100 \text{ times more [1]}}$
(c) **5 percentage points** (21% to 16%) [1]
(d) Any one of: expired air is **warmer**; expired air is **saturated with water vapour [1]**
In (b) the arithmetic $4 \div 0.04$ defeats a lot of students because of the decimal. If the answer is 10 or 1000, the method mark is still there — the error is place value, and it is worth saying so.
- 2** (a) The limewater turns **milky / cloudy**, and quite quickly [1] [3]
(b) Room air contains far less carbon dioxide — about 0.04% against about 4% in expired air [1]; so it takes about a hundred times as much air, and therefore far longer, to produce the same change [1]
The second flask does eventually go cloudy. An answer claiming "nothing happens" has misread the experiment and cannot have the explanation mark.
- 3** **Goblet cells secrete mucus [1]** [3]
which **traps dust, pollen and bacteria** in the air breathed in [1]
Cilia beat and sweep the dirty mucus up the trachea to the throat, where it is swallowed, keeping the airways clear [1]
Both cell types must appear. An answer that only describes mucus has done half the job and gets at most two.
- 4** Breathing in hard lowers the pressure inside the trachea [1] [2]
so without the cartilage rings the tube would be **squashed shut** by the higher pressure outside and air could not reach the lungs [1]
The straw-in-a-thick-milkshake image from the page is a legitimate way to earn the second mark if the physics behind it is stated.
- 5** Any two, one mark each: the nose **warms** the air; the nose **moistens** the air, which [2]
matters because gases diffuse across a moist surface; nose hairs and mucus **filter out dust and bacteria**
Three valid answers exist and only two are needed, so this should be two easy marks. If it is not, the lesson on the airways has not been read.
- 6** (a) $12 \times 500 = 6000 \text{ cm}^3$ per minute [1] = **6 dm³ per minute [1]** [6]
(b) $24 \times 2000 = 48\,000 \text{ cm}^3$ per minute [1] = **48 dm³ per minute [1]**
(c) $48 \div 6 = \mathbf{8 \text{ times greater [1]}}$; both the **rate** (12 → 24 breaths per minute) and the **depth** (500 → 2000 cm³ per breath) increased [1]
Award the first mark of (a) and (b) for the multiplication even if the conversion is missing or wrong — the conversion is the second mark and is where this question is usually lost.

- 7 The muscles **respire faster** to release more energy for contraction **[1]** [4]
 so they use more oxygen and produce **more carbon dioxide**, which raises the carbon dioxide concentration in the blood **[1]**
 This rise is **detected by the brain** **[1]**
 which increases both the **rate and the depth** of breathing, so oxygen is delivered and carbon dioxide removed faster **[1]**
It is the rising carbon dioxide, not the falling oxygen, that triggers the change. An answer built entirely on "the body needs more oxygen" describes the purpose but never gives the mechanism, and tops out at two.
- 8 Count the number of breaths in a **full minute** (or count for 30 s and double it), sitting still **[1]** [3]
Repeat and take a mean to reduce the effect of an unusual reading **[1]**
 Any two kept the same, for one mark: **same time of day; same position** (sitting, not standing); **no exercise beforehand; same room temperature** **[1]**
Watch for the student who counts "in and out" as two breaths. It is one.
- 9 The air is warmed by the body as it passes through the nose and airways, so it leaves at close to body temperature **[1]** [2]
 and it picks up water vapour from the moist lining of the airways and alveoli, so it leaves saturated **[1]**
Both marks are about the air being conditioned on the way through, not about respiration producing water — although the water produced in respiration does contribute, and should not be marked wrong.

Paper C — Hard

30 MARKS

- 1 (a) $3.0 \times 1.6 = 4.8$ and $10^8 \times 10^{-7} = 10^1$ **[1]**, so the total is $4.8 \times 10^1 = 48 \text{ m}^2$ **[1]** [5]
 (b) $48 \div 0.06 = 800$ **times larger** **[1]**
 (c) Diffusion is faster across a larger surface area **[1]**; dividing the lung into millions of tiny sacs multiplies the area available without the chest getting any bigger **[1]**
The powers of ten are the whole difficulty in (a). A student who writes 4.8 without the power has the method and loses only the second mark.
- 2 (a) $1 \times 10^{-6} \text{ m}$ **[1]** [3]
 (b) The diffusion distance is doubled, so oxygen crosses into the blood more slowly **[1]**; less oxygen reaches the blood each minute, so the person becomes breathless and tires quickly, especially when exercising **[1]**
(b) needs a consequence for the person, not only for the molecule. An answer that stops at "diffusion is slower" has answered half the question.
- 3 (a) 0.45 dm^3 **[1]** [6]
 (b) 6 breaths in 30 s, so **12 breaths per minute** **[1]**
 (c) 12×0.45 **[1]** = **5.4 dm³ per minute** **[1]**
 (d) The trace becomes **taller**, showing a greater tidal volume / deeper breaths **[1]**, and the peaks come **closer together**, showing a faster breathing rate **[1]**
In (d) many students give two versions of "he breathes more". Insist on one statement about height and one about spacing — depth and rate are separate measurements and the spirometer shows them separately.

- 4 (a) Rubber sheet = **diaphragm [1]**; glass jar = **rib cage / thorax [1]**; balloons = **lungs [1]** [4]
 (b) Any one: the glass jar **cannot change shape**, whereas real ribs move up and out; there are no **intercostal muscles** in the model; the balloons have **no alveoli**, so no gas exchange takes place; the rubber sheet is **pulled by hand** rather than contracting by itself [1]
Almost everyone gets (a). (b) is the mark that separates, and it is where a student who has actually thought about the model shows it.
- 5 (a) glucose → **lactic acid** (energy released) [1] [3]
 (b) Advantage: it releases energy **without needing oxygen**, so the muscles can keep working when oxygen cannot be delivered fast enough [1]. Disadvantage: it releases **far less energy** per glucose molecule and leaves **lactic acid**, which causes fatigue and pain [1]
Carbon dioxide and water are not products of anaerobic respiration in muscle. Writing them there is the standard error and costs the first mark outright.
- 6 During the sprint her muscles respired anaerobically and built up **lactic acid [1]** [3]
 Extra oxygen is needed to break that lactic acid down — this is the **oxygen debt [1]** so she keeps breathing hard after stopping until the debt has been repaid [1]
"She is out of breath" is a description of what we can see, not an explanation. No marks for it.
- 7 (a) Increase = $40 - 16 = 24$ [1]; $(24 \div 16) \times 100 = 150\%$ [1] [6]
 (b) **Student P [1]**; two reasons from: P has the **lower resting rate** (16 against 20), and P has **almost fully recovered** after three minutes (17 against a resting 16) while Q is still far above resting (30 against 20) [1]
 (c) The muscles built up an **oxygen debt** during the exercise [1]; breathing must stay raised to supply the extra oxygen needed to remove the lactic acid before it can return to normal [1]
In (a), dividing by 40 instead of 16 gives 60% — measuring the change against the value you finished on rather than the one you started from. Award the first mark only. This is the same trap as the percentage change work in the maths papers, in a biology costume.

Paper D — Hard

30 MARKS

- 1 (a) Fall = $50 - 20 = 30 \text{ m}^2$ [1]; $(30 \div 50) \times 100 = 60\%$ [1] [5]
 (b) The surface area for gas exchange is much smaller, so **less oxygen diffuses into the blood each minute [1]**; his muscles cannot get enough oxygen for aerobic respiration [1]; so he becomes **breathless and has to stop part-way up the stairs [1]**
Dividing by 20 rather than 50 gives 150% — an impossible answer, since a quantity cannot fall by more than 100%. Point that out rather than just marking it wrong; the check is free and he can do it himself next time.
- 2 (a) Carbon monoxide binds to the haemoglobin and does not release it [1], so fewer haemoglobin molecules are left free to carry oxygen and the blood carries less oxygen even though the lungs are working normally [1] [4]
 (b) 12% of $250 = 30 \text{ cm}^3$ [1]; $250 - 30 = 220 \text{ cm}^3$ per minute [1]
Accept 250×0.88 in one step for both marks. An answer of 30 has calculated the loss and forgotten to subtract it — one mark.

- 3** **Mucus is no longer swept out of the airways**, so it collects in the lungs **[1]** [3]
 The smoker **coughs repeatedly** to shift it — smoker's cough **[1]**
 Trapped dust and bacteria stay in the lungs, so **chest infections become more frequent** **[1]**
Any two clear consequences with explanations reach three marks. A bare list without reasons gets one.
- 4** Into the **nose**, where the air is warmed, moistened and filtered **[1]** [6]
 Down the **trachea**, into a **bronchus**, then a **bronchiole**, and into an **alveolus**; it moves because the diaphragm and intercostals contracted, increasing the volume of the thorax and lowering the pressure inside the lungs **[2]**
 Across the **alveolus wall and the capillary wall** — about 1 µm of tissue, both one cell thick — by **diffusion**, because oxygen is at a higher concentration in the alveolar air than in the blood **[1]**
 It binds to **haemoglobin** in a red blood cell and is carried in the blood, pumped by the heart, to the leg muscle **[1]**
 It **diffuses out of the blood into the muscle cell**, down the concentration gradient, because the respiring cell keeps using oxygen up and so keeps its concentration low **[1]**
This is the question that shows whether the topic has been joined up or learnt in pieces. Diffusion must be named at both ends, and a gradient must be explained at least once. An answer that lists the structures correctly but never says why the oxygen moves cannot go above three.
- 5** (a) Each breath contains fewer oxygen molecules, so breathing faster and more deeply moves **more air, and therefore more oxygen, into the lungs each minute** **[1]**; this keeps the concentration gradient between alveolar air and blood as large as possible **[1]** [4]
 (b) More red blood cells means more **haemoglobin** **[1]**, so the blood can carry more oxygen from each breath to the respiring cells **[1]**
Nothing here has been taught directly — it is the alveolus material applied to an unfamiliar situation. Credit any answer that reaches the right idea by a different route.
- 6** Anaerobic, oxygen needed: **no** [3]
 Anaerobic, products: **lactic acid**
 Aerobic, energy released: **much more** (per glucose molecule)
One mark per box.
- 7** (i) The mistakes: respiration does **not** take place in the lungs, and oxygen is **not** turned into carbon dioxide **[1]** [5]
 Correct version: respiration takes place in **every cell of the body**, in the mitochondria; the lungs carry out gas exchange **[1]**. The carbon in the carbon dioxide comes from the **glucose**, not from the oxygen — oxygen is a reactant, not raw material that is converted **[1]**
 (ii) The mistake: during exhalation the diaphragm **relaxes and moves up**; it does not push down, and muscles cannot push air at all **[1]**
 Correct version: the diaphragm relaxes and domes upwards and the ribs move down and in, so the **volume of the thorax decreases**, the **pressure inside rises above atmospheric**, and air flows out down the pressure gradient **[1]**
These are the two sentences this whole topic is built to prevent. Sentence (i) is the breathing / respiration confusion plus the "oxygen turns into carbon dioxide" error; sentence (ii) skips the volume and pressure steps entirely. Marking them in someone else's work is the same cognitive job as not writing them, and far easier to face.